

Multiple Regression

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MZR - Spring 2023 - STA201

Multiple Regression

- ▶ Multiple regression models are similar to simple linear regression models except they include more than one independent variables.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

- ▶ Where,
- ▶ Y= dependent variable
- ▶ X= independent variable
- ▶ β_0 = Intercept
- ▶ β_j = Slope; $j=1, 2, \dots, n$
- ▶ ϵ = Error term (unexplained factor)
- ▶ The multiple regression analysis finds the coefficients for each independent variable so that they make the line that has the lowest sum of squared errors.

} Regression coefficients
(Parameters)

Interpretation of Parameters in Multiple Regression

- A regression coefficient in multiple regression is the slope of the linear relationship between the response variable and the part of a predictor variable that is independent of all other predictor variables.
- β_j represents the change in the response variable(Y) associated with a change of one in the predictor variable(X_j) when all other predictor variables are held constant.

Assumptions of Multiple Regression

- The relationship between the dependent variable and the set of independent variables must be linear.
- The residuals follow the normal probability distribution.
- The independent variables are not correlated.
- Residuals are independent of each other.

Example

- ▶ For three independent variables designated X_1 , X_2 , and X_3 , the general multiple regression equation is :

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

- ▶ A Real Estate sells homes in the capital city. One of the questions most frequently asked by prospective buyers is: If we purchase this home, how much can we expect to pay to heat it during the winter? Three variables are thought to relate to the heating costs:
 - ▶ the mean daily outside temperature,
 - ▶ the number of inches of insulation in the attic, and
 - ▶ the age of the furnace

Example

- ▶ To investigate, they have selected a random sample of 20 recently sold homes. They determined the cost to heat the home last January, as well as the January outside temperature, the number of inches of insulation in the attic, and the age of the furnace. The sample information is given in the following table:

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Home	Heating Cost (\$)	Mean Outside temperature (°F)	Attic Insulation (inches)	Age of furnace (years)
1	250	35	3	6
2	360	29	4	10
3	165	36	7	3
4	43	60	6	9
5	92	65	5	6
6	200	30	5	5
7	355	10	6	7
8	290	7	10	10
9	230	21	9	11
10	120	55	2	5
11	73	54	12	4
12	205	48	5	1
13	400	20	5	15
14	230	39	4	7
15	72	60	8	6
16	272	20	5	8
17	94	58	7	3
18	190	40	8	11
19	235	27	9	8
20	139	30	7	5

Example

- ▶ Determine the multiple regression equation. Which variables are the independent variables? Which variable is the dependent variable? Discuss the regression coefficients. What does it indicate that some are positive and some are negative? What is the intercept value?
- ▶ What is the estimated heating cost for a home if the mean outside temperature is 30 degrees, there are 5 inches of insulation in the attic, and the furnace is 10 years old?

Example

- ▶ Solution:
- ▶ The dependent variable is the January heating cost. There are three independent variables: the mean outside temperature, the number of inches of insulation in the attic, and the age of the furnace.
- ▶ The general form of a multiple regression equation with three independent variables is:
- ▶ $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$
- ▶ Using Statistical software the multiple regression equation is:
- ▶ $Y = 427 - 4.58X_1 - 14.8X_2 + 6.10X_3$
- ▶ The intercept value is 427. This is the point where the regression equation crosses the Y – axis

Example

- ▶ The regression coefficients for the mean outside temperature and the amount of attic insulation are both negative. As the outside temperature increases, the cost to heat the home will go down. It is an inverse relationship. For each degree the mean temperature increases, we expect the heating cost decreases \$4.58 per month.
- ▶ The variable “attic insulation” also shows an inverse relationship: the more insulation in the attic the less the cost to heat the home. So the negative sign for this coefficient is logical. For each additional inch of insulation, we expect the cost to heat the home to decline \$14.80 per month, regardless of the outside temperature or the age of the furnace.

Example

- ▶ The furnace variable shows a direct relationship. For each additional year old the furnace is, we expect the cost to increase \$6.10 per month.
- ▶ If the mean outside temperature for the month is 30 degrees, there are 5 inches of insulation in the attic, and the furnace is 10 years old, the estimated heating cost for the month :
- ▶ $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 = 427 - 4.58X_1 - 14.8X_2 + 6.10X_3$
 $= 427 - 4.58(30) - 14.8(5) + 6.10(10) = 276.60.$

Example: R- Output

```
Console Terminal x Jobs x
~/
Residuals:
      Min       1Q   Median       3Q      Max
-3.4080 -0.7097 -0.0078  0.7167  3.1679

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -11.747065   0.476899  -24.632  < 2e-16 ***
Age           0.126368   0.017851   7.079  3.45e-12 ***
Height        0.278432   0.009926  28.051  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 722 degrees of freedom
Multiple R-squared:  0.843,    Adjusted R-squared:  0.8425
F-statistic: 1938 on 2 and 722 DF,  p-value: < 2.2e-16
```

Example: Fitted Model

Console Terminal x Jobs x

~/

Residuals:

	Min	1Q	Median	3Q
	-3.4080	-0.7097	-0.0078	0.7167

Coefficients:

	Estimate	Std. Error			
(Intercept)	-11.747065	0.47689			
Age	0.126368	0.017851	7.079	3.45e-12	***
Height	0.278432	0.009926	28.051	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 722 degrees of freedom
Multiple R-squared: 0.843, Adjusted R-squared: 0.8425
F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16

LungCap=-11.75+0.13*Age+0.28*Height

Example: Interpretation

```
Console Terminal x Jobs x
~/
Residuals:
    Min       1Q   Median       3Q      Max
-3.4080 -0.7097 -0.0078  0.7097  3.4080

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -11.747065    0.126368  -92.99 < 2e-16 ***
Age           0.126368    0.005526   22.87 < 2e-16 ***
Height        0.278432    0.005526   50.39 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 722 degrees of freedom
Multiple R-squared:  0.843,    Adjusted R-squared:  0.8425
F-statistic: 1938 on 2 and 722 DF,  p-value: < 2.2e-16
```

LungCap=-11.75+0.13*Age+0.28*Height

On an average lung capacity of a person in youth age is -11.75 liters when other variables (age and height) are set to 0 which is meaningless. Because height of a person can't be 0.

Example: Interpretation

Console Terminal x Jobs x

~/

Residuals:

	Min	1Q	Median	0.
	-3.4080	-0.7097	-0.0078	0.

Coefficients:

	Estimate	Std.
(Intercept)	-11.747065	0.
Age	0.126368	0.
Height	0.278432	0.009920

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 722 degrees of freedom
Multiple R-squared: 0.843, Adjusted R-squared: 0.8425
F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16

$$\text{LungCap} = -11.75 + 0.13 * \text{Age} + 0.28 * \text{Height}$$

When height remains constant on an average an increase in age of one year, the estimated lung capacity increases by 0.13 liter in youth age.

Example: Interpretation

```
Console Terminal x Jobs x
~/
Residuals:
      Min       1Q   Median       3Q      Max
-3.4080 -0.7097 -0.0078  0.7100  3.4080

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -11.747065    0.41000  -28.641 < 2e-16 ***
Age           0.126368    0.01000   12.637 < 2e-16 ***
Height       0.278432    0.009926  28.051 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.056 on 722 degrees of freedom
Multiple R-squared:  0.843,    Adjusted R-squared:  0.8425
F-statistic: 1938 on 2 and 722 DF,  p-value: < 2.2e-16
```

LungCap=-11.75+0.13*Age+0.28*Height

When age remains constant on an average an increase in height of one inch, the estimated lung capacity increases by 0.28 liter in youth age.

Example: Interpretation

Console				
Terminal x Jobs x				
~/				
Residuals:				
Min	1Q	Median	3Q	Max
-3.4080	-0.7097	-0.0078	0.7167	3.1
Coefficients:				
	Estimate	Std. Error	t	Pr > t
(Intercept)	-11.747065	0.476899	-24.632	< 2e-16 ***
Age	0.126368	0.017851	7.079	3.45e-12 ***
Height	0.278432	0.009926	28.051	< 2e-16 ***
--- Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				
Residual standard error: 1.056 on 722 degrees of freedom				
Multiple R-squared: 0.843, Adjusted R-squared: 0.8425				
F-statistic: 1938 on 2 and 722 DF, p-value: < 2.2e-16				

84% of the observed variability in the lung capacity in youth age is explained by the model

Multiple regression Example

- More than one independent variable
 - Son's height dependent on:
 - Father's height
 - Father's age
 - Socio-economic status (SES)
 - Mother's BMI (body mass index)
 - Son's physical exercise

Multiple Regression

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- ▶ Numeric Response variable (Y)
- ▶ p Numeric predictor variables
- ▶ Model:
$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \varepsilon$$
- ▶ Partial Regression Coefficients: β_i = effect (on the mean response) of changing the i^{th} predictor variable by 1 unit, **holding all other predictors constant**

Regression Output

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Descriptive Statistics

Correlations

Variables Entered/ Removed

Model summary

ANOVA

Coefficients

Residuals

Histogram and P-P plot